

CSE 604

Artificial Intelligence

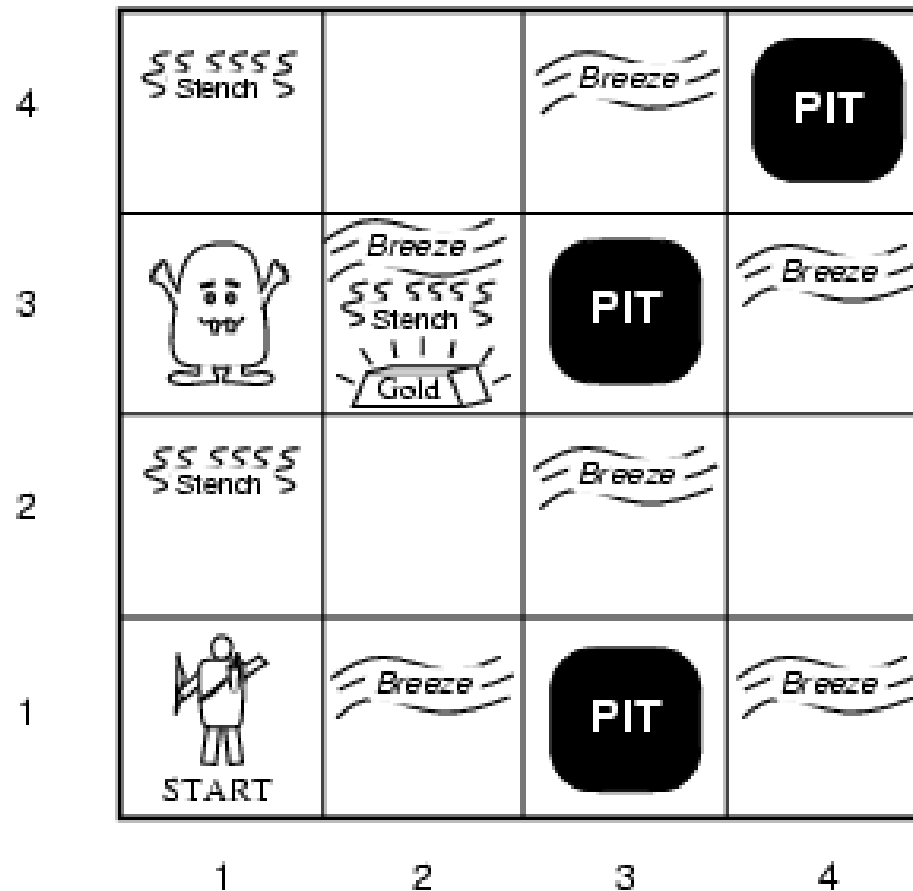
Chapter 7: Logical Agents

Adapted from slides available in Russell & Norvig's textbook webpage

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Wumpus world



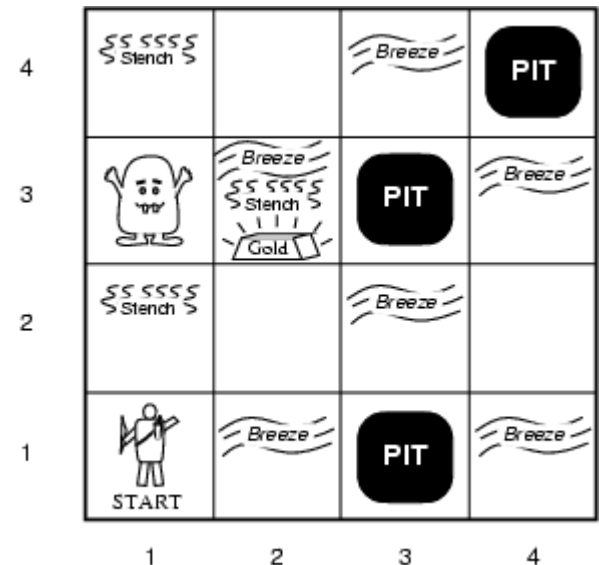
Wumpus World PEAS description

- Performance measure

- gold +1000, death -1000
- -1 per step, -10 for using the arrow

- Environment

- Squares adjacent to wumpus are smelly
- Squares adjacent to pit are breezy
- Glitter iff gold is in the same square
- Shooting kills wumpus if you are facing it
- Shooting uses up the only arrow
- Grabbing picks up gold if in same square
- Releasing drops the gold in same square

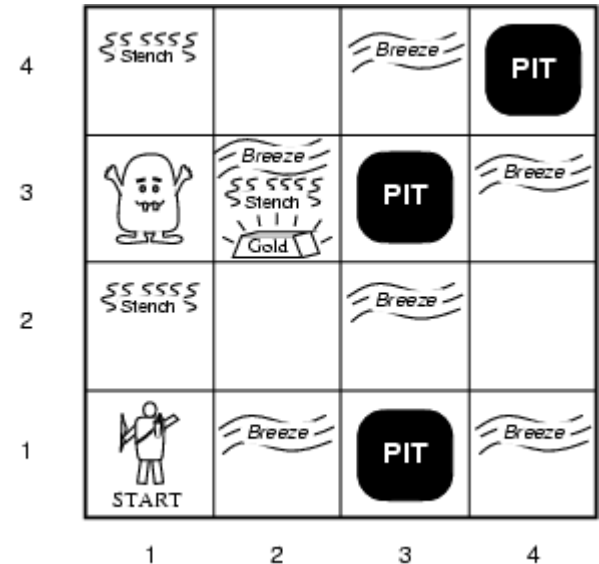


- Sensors: Stench, Breeze, Glitter, Bump, Scream

- Actuators: Left turn, Right turn, Forward, Grab, Release, Shoot

Wumpus World Environment

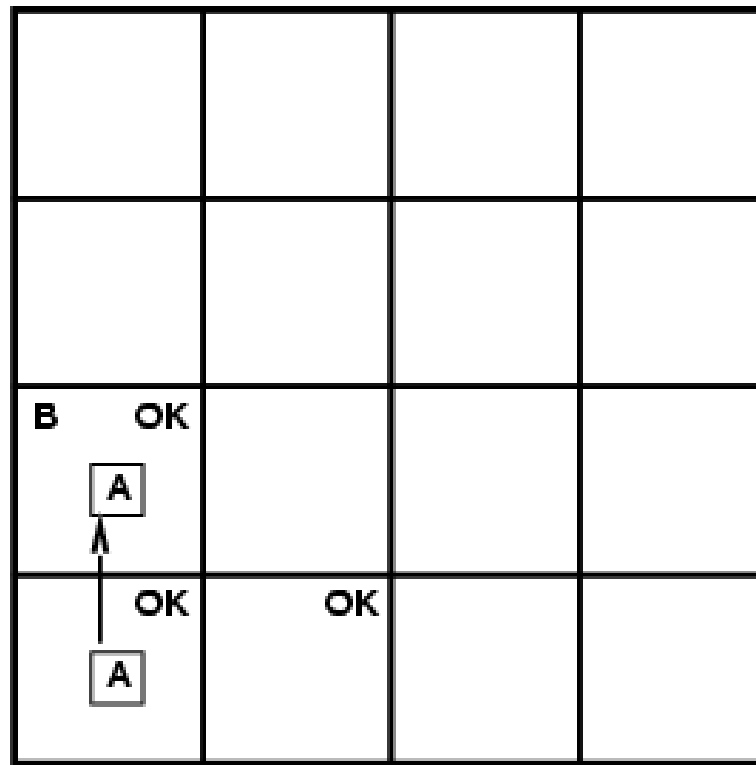
- Partially Observable: only local perception
- Single-agent Wumpus is essentially a natural feature
- Deterministic outcomes exactly specified
- Sequential: rewards may come after many actions
- Static: Wumpus and Pits do not move
- Discrete: One state at a time



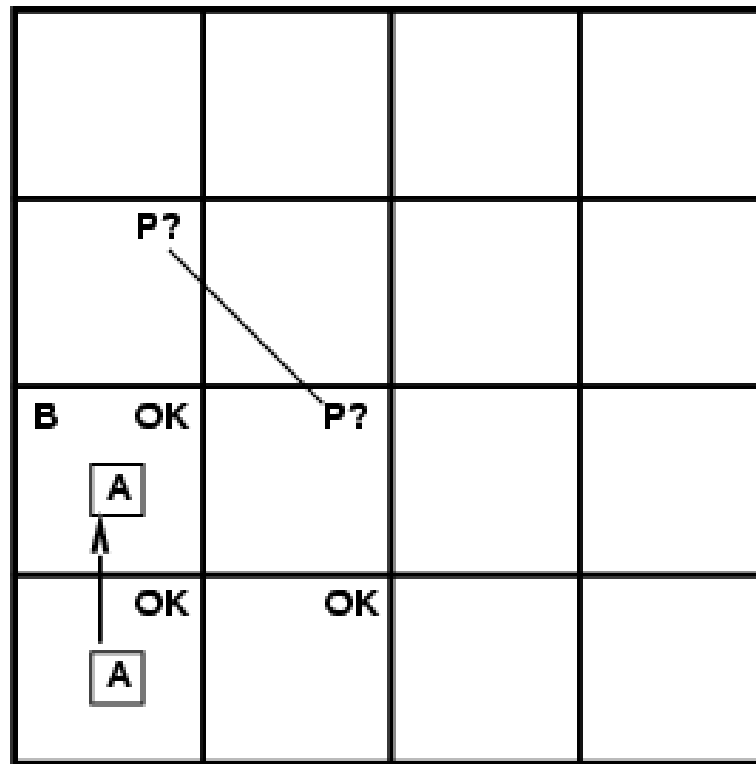
Exploring a wumpus world

OK			
OK A	OK		

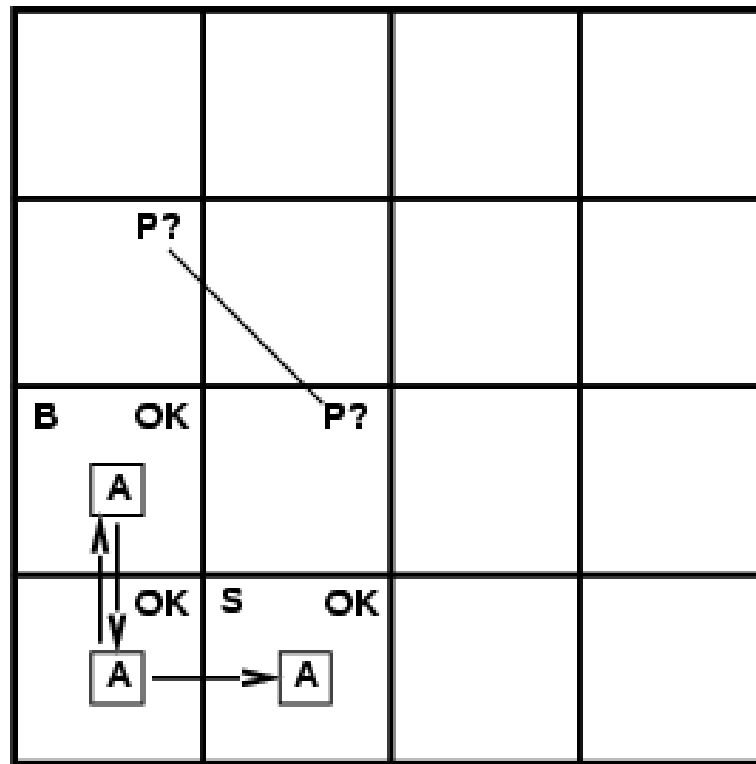
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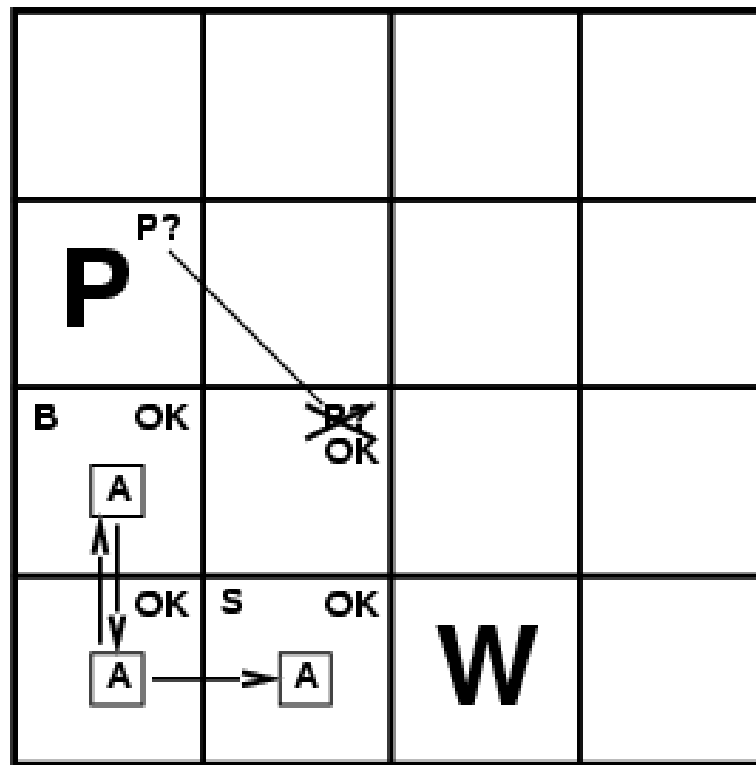
Exploring a wumpus world



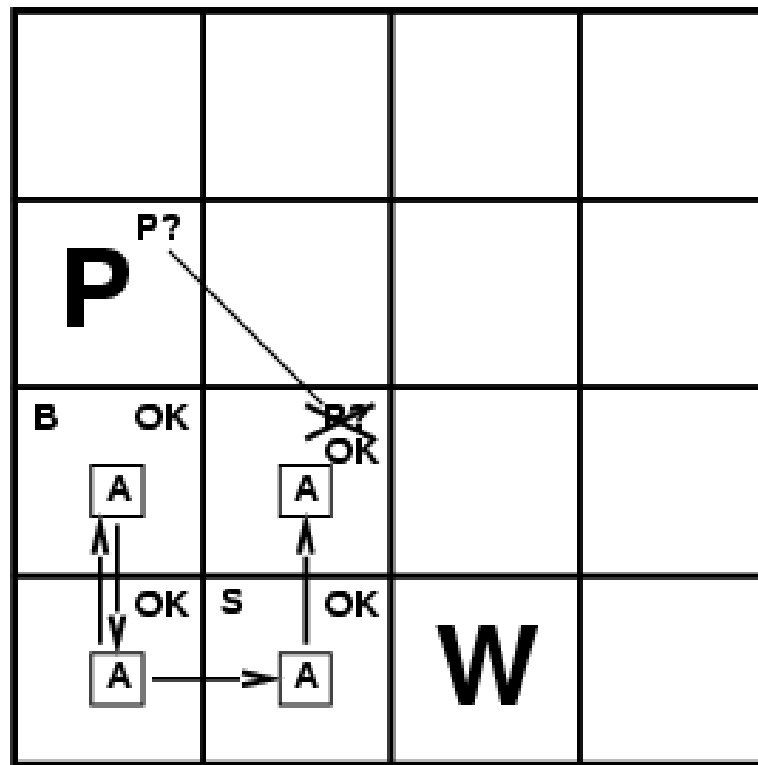
Exploring a wumpus world



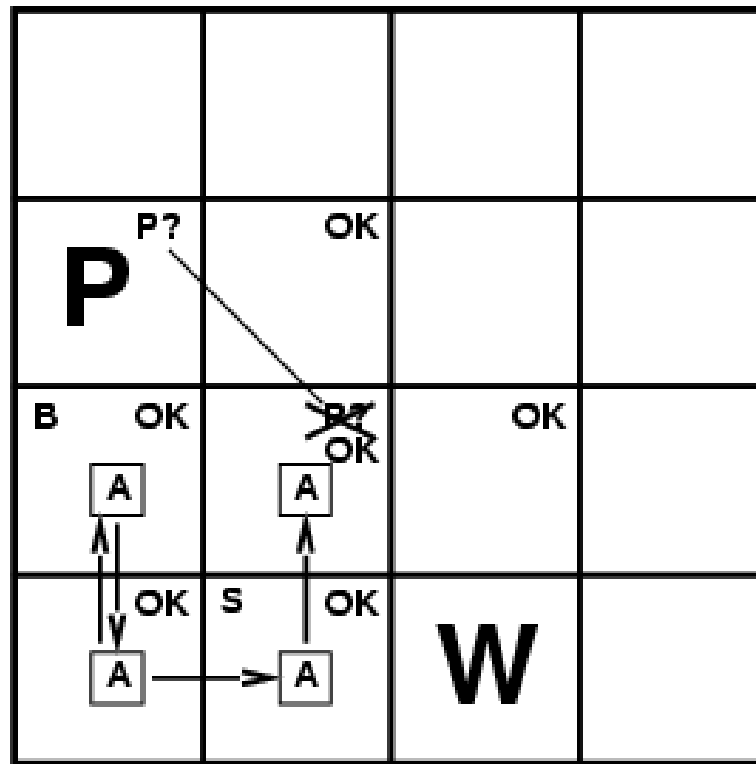
Exploring a wumpus world



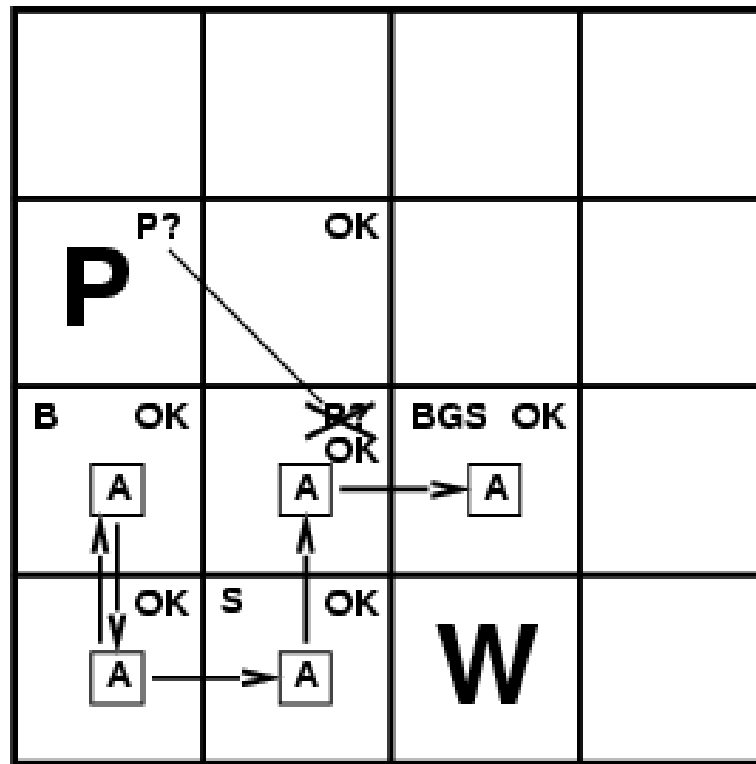
Exploring a wumpus world



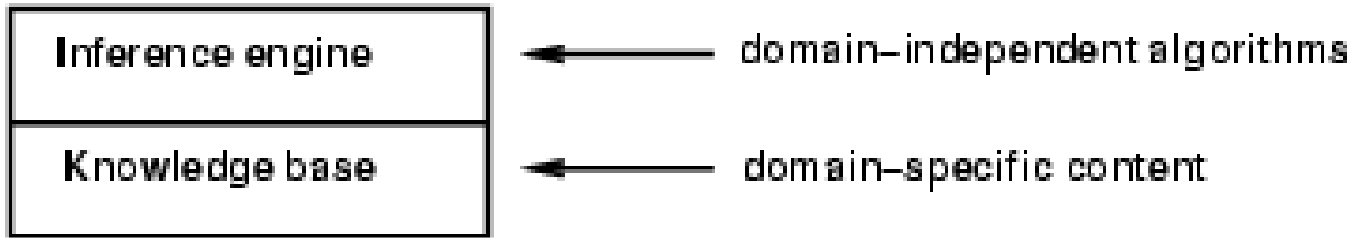
Exploring a wumpus world



Exploring a wumpus world



Knowledge bases



- Knowledge base = set of **sentences** in a **formal** language
- **Declarative** approach to building an agent (or other system):
 - Tell it what it needs to know
- Then it can Ask itself what to do - answers should follow from the KB
- Agents can be viewed at the **knowledge level**
i.e., what they know, regardless of how implemented
- Or at the **implementation level**
 - i.e., data structures in KB and algorithms that manipulate them

A simple knowledge-based agent

```
function KB-AGENT(percept) returns an action  
  static: KB, a knowledge base  
           t, a counter, initially 0, indicating time  
  
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))  
  action ← ASK(KB, MAKE-ACTION-QUERY(t))  
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))  
  t ← t + 1  
  return action
```

- The agent must be able to:
 - Represent states, actions, etc.
 - Incorporate new percepts
 - Update internal representations of the world
 - Deduce hidden properties of the world
 - Deduce appropriate actions

Logic in general

- **Logics** are formal languages for representing information such that conclusions can be drawn
- **Syntax** defines the sentences in the language
- **Semantics** define the "meaning" of sentences;
 - i.e., define **truth** of a sentence in a world
- E.g., the language of arithmetic
 - $x+2 \geq y$ is a sentence; $x^2+y > \{ \}$ is not a sentence
 - $x+2 \geq y$ is true iff the number $x+2$ is no less than the number y
 - $x+2 \geq y$ is true in a world where $x = 7, y = 1$
 - $x+2 \geq y$ is false in a world where $x = 0, y = 6$

Entailment

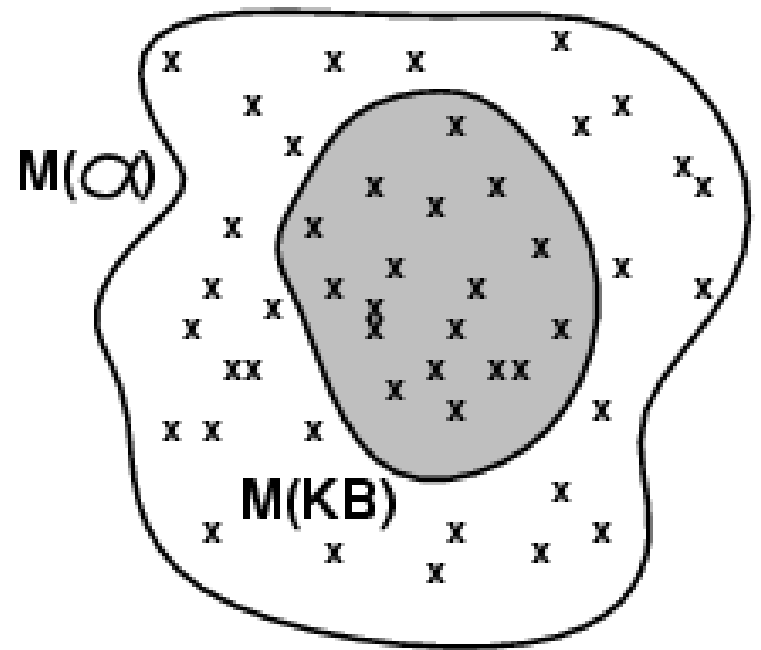
- Entailment means that one thing follows from another:

$$KB \models \alpha$$

- Knowledge base KB entails sentence α if and only if α is true in all worlds where KB is true
 - E.g., $x = 0$ entails $xy = 0$
 - Entailment is a relationship between sentences (i.e., syntax) that is based on semantics

Models

- Logicians typically think in terms of **models**, which are formally structured worlds with respect to which truth can be evaluated
- We say m **is a model of** a sentence α if α is true in m
- $M(\alpha)$ is the set of all models of α
- Then $KB \models \alpha$ iff $M(KB) \subseteq M(\alpha)$
 - E.g. $KB = \text{Giants won and Reds won}$
 $\alpha = \text{Giants won}$

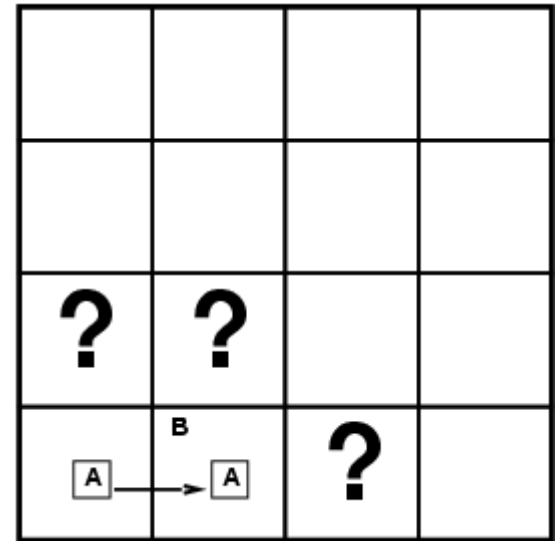


Entailment in the wumpus world

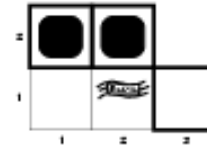
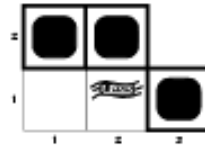
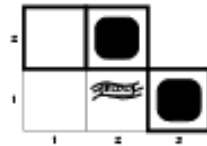
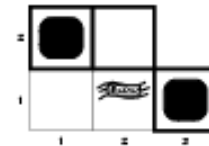
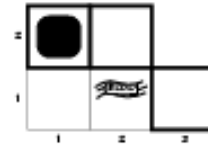
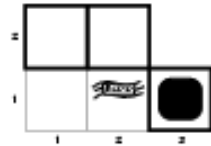
Situation after detecting nothing in
[1,1], moving right, breeze in [2,1]

Consider possible models for *KB*
assuming only pits

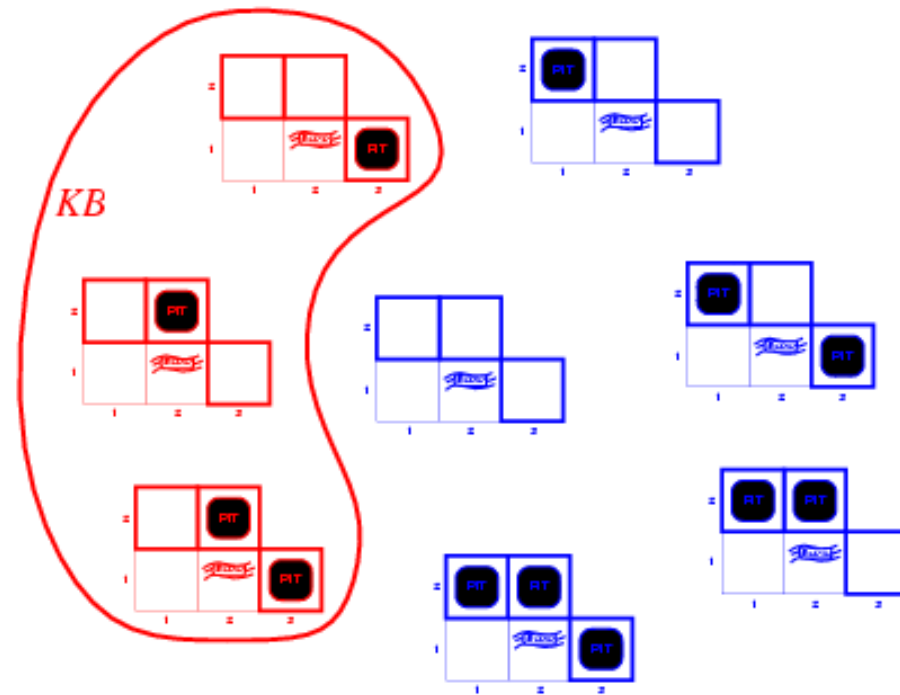
3 Boolean choices $\Rightarrow$ 8 possible
models



Wumpus models

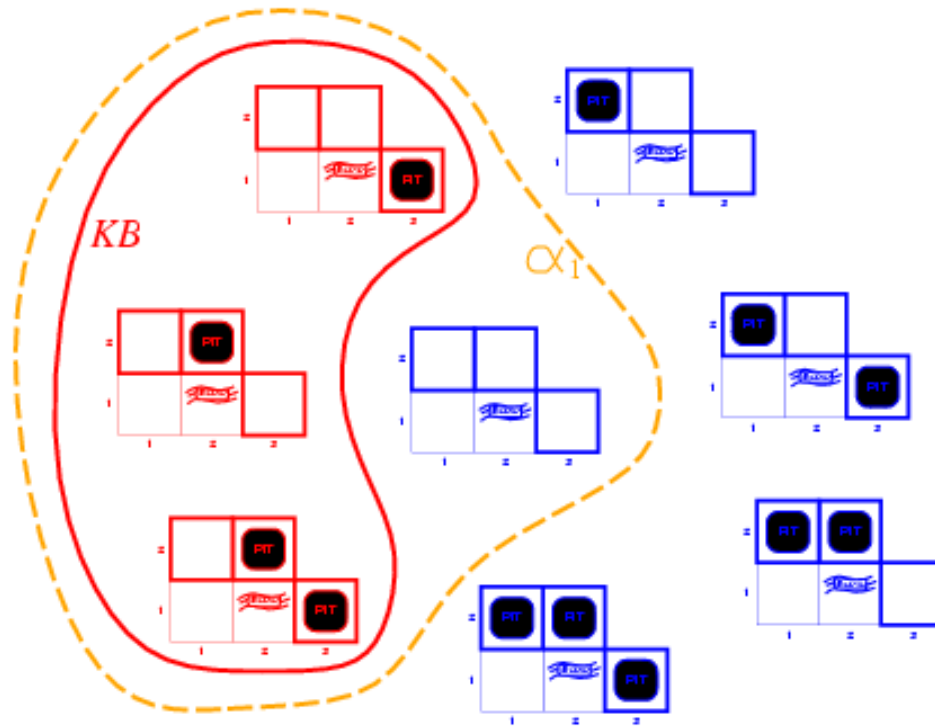


Wumpus models



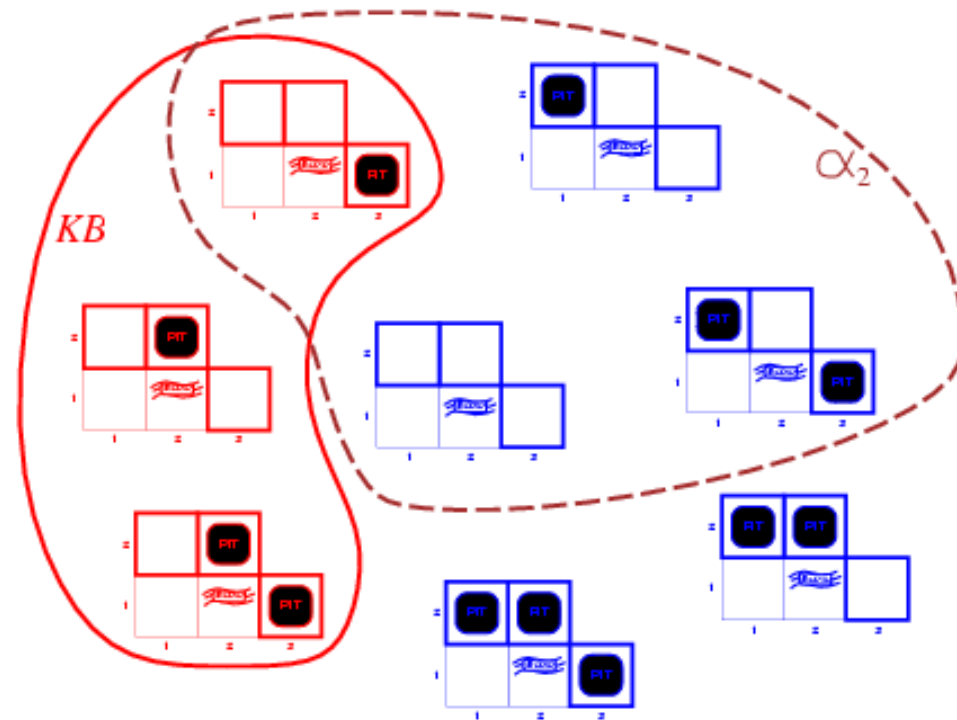
- $KB = \text{wumpus-world rules} + \text{observations}$

Wumpus models



- KB = wumpus-world rules + observations
- $\alpha_1 = "[1,2] \text{ is safe}]", KB \models \alpha_1$, proved by model checking

Wumpus models



- KB = wumpus-world rules + observations
- α_2 = "[2,2] is safe", $KB \not\models \alpha_2$

Inference

- $KB \vdash_i \alpha$ = sentence α can be derived from KB by procedure i
- **Soundness**: i is sound if whenever $KB \vdash_i \alpha$, it is also true that $KB \models \alpha$
- **Completeness**: i is complete if whenever $KB \models \alpha$, it is also true that $KB \vdash_i \alpha$
- Preview: we will define a logic (first-order logic) which is expressive enough to say almost anything of interest, and for which there exists a sound and complete inference procedure.
- That is, the procedure will answer any question whose answer follows from what is known by the KB .

Propositional logic: Syntax

- Propositional logic is the simplest logic – illustrates basic ideas
- The proposition symbols P_1, P_2 etc are sentences
 - If S is a sentence, $\neg S$ is a sentence (negation)
 - If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence (conjunction)
 - If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence (disjunction)
 - If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence (implication)
 - If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence (biconditional)

Propositional logic: Semantics

Each model specifies true/false for each proposition symbol

E.g. $P_{1,2}$ $P_{2,2}$ $P_{3,1}$
false true false

With these symbols, 8 possible models, can be enumerated automatically.

Rules for evaluating truth with respect to a model m :

$\neg S$ is true iff S is false

$S_1 \wedge S_2$ is true iff S_1 is true and S_2 is true

$S_1 \vee S_2$ is true iff S_1 is true or S_2 is true

$S_1 \Rightarrow S_2$ is true iff S_1 is false or S_2 is true

i.e., is false iff S_1 is true and S_2 is false

$S_1 \Leftrightarrow S_2$ is true iff $S_1 \Rightarrow S_2$ is true and $S_2 \Rightarrow S_1$ is true

Simple recursive process evaluates an arbitrary sentence, e.g.,

$$\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \text{true} \wedge (\text{true} \vee \text{false}) = \text{true} \wedge \text{true} = \text{true}$$

Truth tables for connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

Wumpus world sentences

Let $P_{i,j}$ be true if there is a pit in $[i, j]$.

Let $B_{i,j}$ be true if there is a breeze in $[i, j]$.

$$\neg P_{1,1}$$

$$\neg B_{1,1}$$

$$B_{2,1}$$

- "Pits cause breezes in adjacent squares"

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

Truth tables for inference

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	KB	α_1
<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>true</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<u><i>true</i></u>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<u><i>true</i></u>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>	<u><i>true</i></u>	<u><i>true</i></u>
<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>false</i>

Inference by enumeration

- Depth-first enumeration of all models is sound and complete

```
function TT-ENTAILS?(KB,  $\alpha$ ) returns true or false  
    symbols  $\leftarrow$  a list of the proposition symbols in KB and  $\alpha$   
    return TT-CHECK-ALL(KB,  $\alpha$ , symbols, [])  
  
function TT-CHECK-ALL(KB,  $\alpha$ , symbols, model) returns true or false  
    if EMPTY?(symbols) then  
        if PL-TRUE?(KB, model) then return PL-TRUE?( $\alpha$ , model)  
        else return true  
    else do  
        P  $\leftarrow$  FIRST(symbols); rest  $\leftarrow$  REST(symbols)  
        return TT-CHECK-ALL(KB,  $\alpha$ , rest, EXTEND(P, true, model)) and  
            TT-CHECK-ALL(KB,  $\alpha$ , rest, EXTEND(P, false, model))
```

- For n symbols, time complexity is $O(2^n)$, space complexity is $O(n)$

Logical equivalence

- Two sentences are **logically equivalent** iff true in same models: $\alpha \equiv \beta$ iff $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

Validity and satisfiability

A sentence is **valid** if it is true in **all** models,

e.g., *True*, $A \vee \neg A$, $A \Rightarrow A$, $(A \wedge (A \Rightarrow B)) \Rightarrow B$

Validity is connected to inference via the **Deduction Theorem**:

$KB \models \alpha$ if and only if $(KB \Rightarrow \alpha)$ is valid

A sentence is **satisfiable** if it is true in **some** model

e.g., $A \vee B$, C

A sentence is **unsatisfiable** if it is true in **no** models

e.g., $A \wedge \neg A$

Satisfiability is connected to inference via the following:

$KB \models \alpha$ if and only if $(KB \wedge \neg \alpha)$ is unsatisfiable

Proof methods

- Proof methods divide into (roughly) two kinds:
 - Application of inference rules
 - Legitimate (sound) generation of new sentences from old
 - **Proof** = a sequence of inference rule applications
 - Can use inference rules as operators in a standard search algorithm
 - Typically require transformation of sentences into a **normal form**
 - Model checking
 - truth table enumeration (always exponential in n)
 - improved backtracking, e.g., Davis--Putnam-Logemann-Loveland (DPLL)
 - heuristic search in model space (sound but incomplete)
 - e.g., min-conflicts-like hill-climbing algorithms

Resolution

Conjunctive Normal Form (CNF)

conjunction of disjunctions of literals

E.g., $(A \vee \neg B) \wedge (B \vee \neg C \vee \neg D)$

- Resolution inference rule (for CNF):

$$\ell_i \vee \dots \vee \ell_k,$$

$$m_1 \vee \dots \vee m_n$$

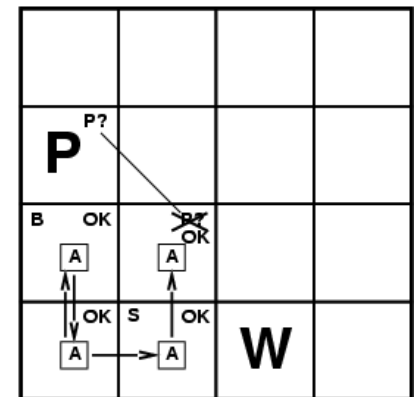
$$\ell_i \vee \dots \vee \ell_{i-1} \vee \ell_{i+1} \vee \dots \vee \ell_k \vee m_1 \vee \dots \vee m_{j-1} \vee m_{j+1} \vee \dots \vee m_n$$

where ℓ_i and m_j are complementary literals, E.g.,

$$P_{1,3} \vee P_{2,2}, \quad \neg P_{2,2}$$

$$P_{1,3}$$

- Resolution is sound and complete for propositional logic



Conversion to CNF

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

1. Eliminate $\Leftrightarrow$, replacing $\alpha \Leftrightarrow \beta$ with $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$.

$$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

2. Eliminate $\Rightarrow$, replacing $\alpha \Rightarrow \beta$ with $\neg\alpha \vee \beta$.

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1})$$

3. Move $\neg$ inwards using de Morgan's rules and double-negation:

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1})$$

4. Apply distributivity law ($\wedge$ over $\vee$) and flatten:

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$$

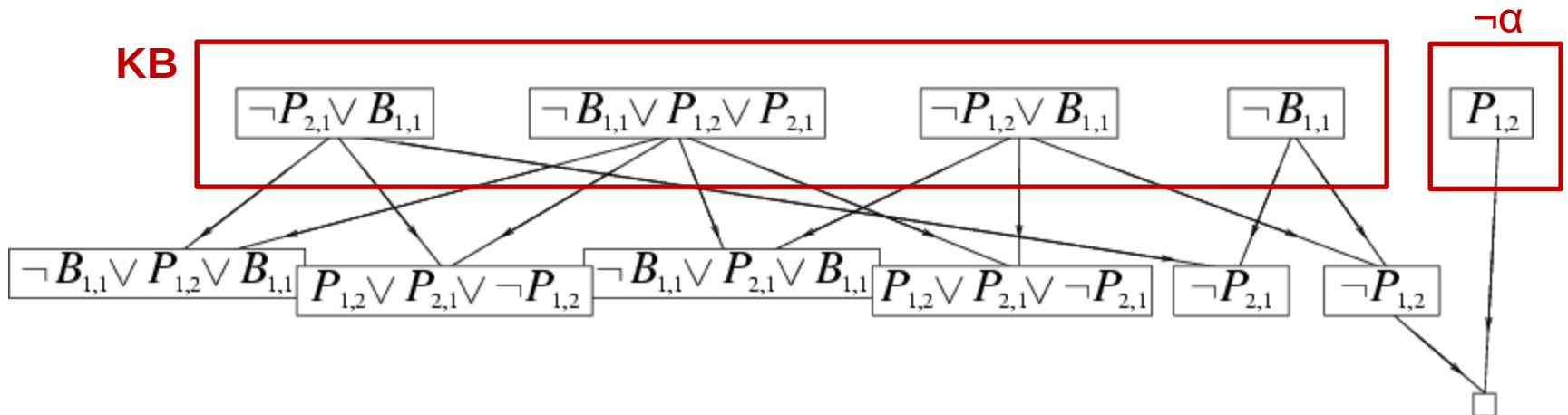
Resolution algorithm

- Proof by contradiction, i.e., show $KB \wedge \neg \alpha$ unsatisfiable

```
function PL-RESOLUTION( $KB, \alpha$ ) returns true or false  
   $clauses \leftarrow$  the set of clauses in the CNF representation of  $KB \wedge \neg \alpha$   
   $new \leftarrow \{ \}$   
  loop do  
    for each  $C_i, C_j$  in  $clauses$  do  
       $resolvents \leftarrow$  PL-RESOLVE( $C_i, C_j$ )  
      if  $resolvents$  contains the empty clause then return true  
       $new \leftarrow new \cup resolvents$   
  if  $new \subseteq clauses$  then return false  
   $clauses \leftarrow clauses \cup new$ 
```

Resolution example

- $KB = (B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})) \wedge \neg B_{1,1} \quad \alpha = \neg P_{1,2}$



In-class Example

If the unicorn is mythical, then it is immortal, but if it is not mythical, then it is a mortal mammal. If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned.

Prove that the unicorn is both magical and horned.

(Adapted from Barwise and Etchemendy, 1993.)

Forward and backward chaining

- Horn Form (restricted)

KB = conjunction of Horn clauses

- Horn clause = Clause with at most one positive literal
 - proposition symbol; or
 - (conjunction of symbols) $\Rightarrow$ symbol
- E.g., $C \wedge (B \Rightarrow A) \wedge (C \wedge D \Rightarrow B)$

- Modus Ponens (for Horn Form): complete for Horn KBs

$$\frac{\alpha_1, \dots, \alpha_n, \quad \alpha_1 \wedge \dots \wedge \alpha_n \Rightarrow \beta}{\beta}$$

- Can be used with forward chaining or backward chaining.
- These algorithms are very natural and run in linear time

Forward chaining

- **Idea:** fire any rule whose premises are satisfied in the *KB*,
 - add its conclusion to the *KB*, until query is found

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

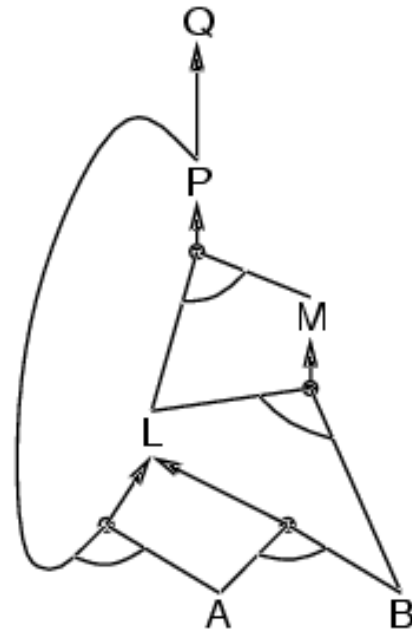
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining algorithm

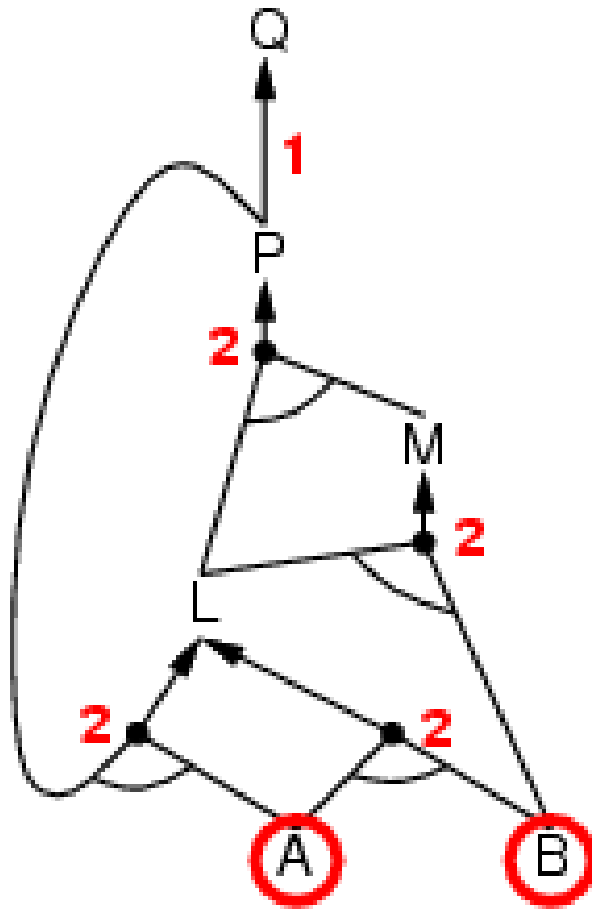
```
function PL-FC-ENTAILS?(KB, q) returns true or false
  local variables: count, a table, indexed by clause, initially the number of premises
                  inferred, a table, indexed by symbol, each entry initially false
                  agenda, a list of symbols, initially the symbols known to be true

  while agenda is not empty do
    p ← POP(agenda)
    unless inferred[p] do
      inferred[p] ← true
      for each Horn clause c in whose premise p appears do
        decrement count[c]
        if count[c] = 0 then do
          if HEAD[c] = q then return true
          PUSH(HEAD[c], agenda)

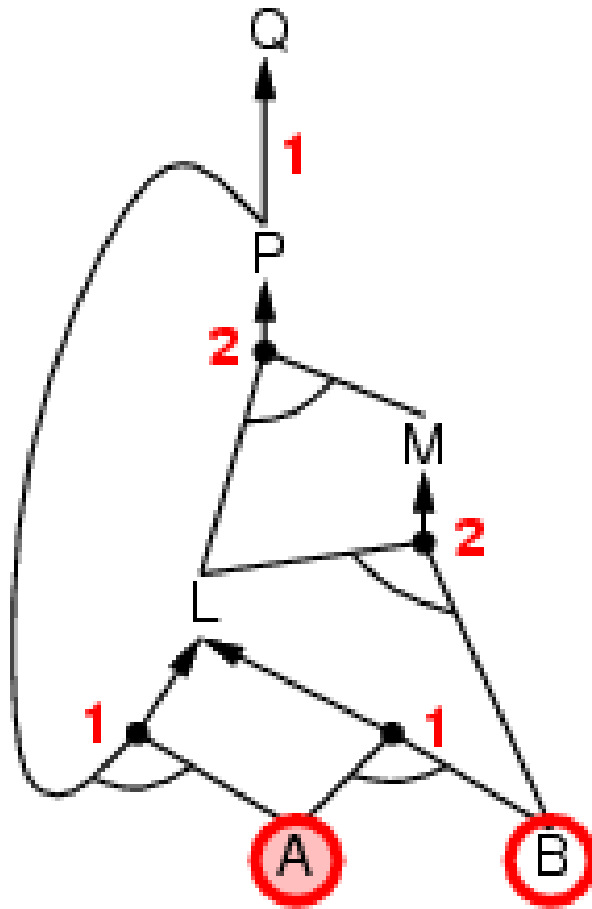
  return false
```

- Forward chaining is sound and complete for Horn KB

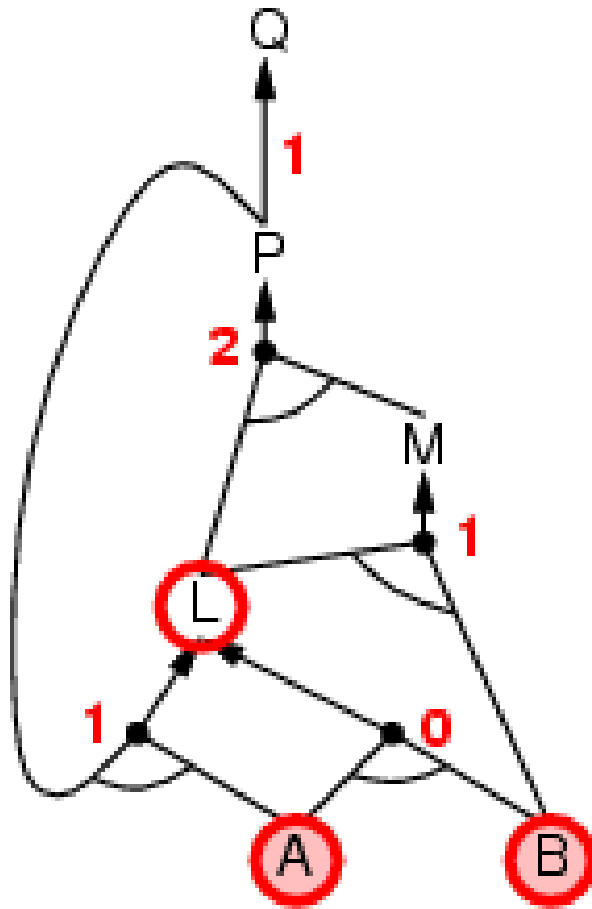
Forward chaining example



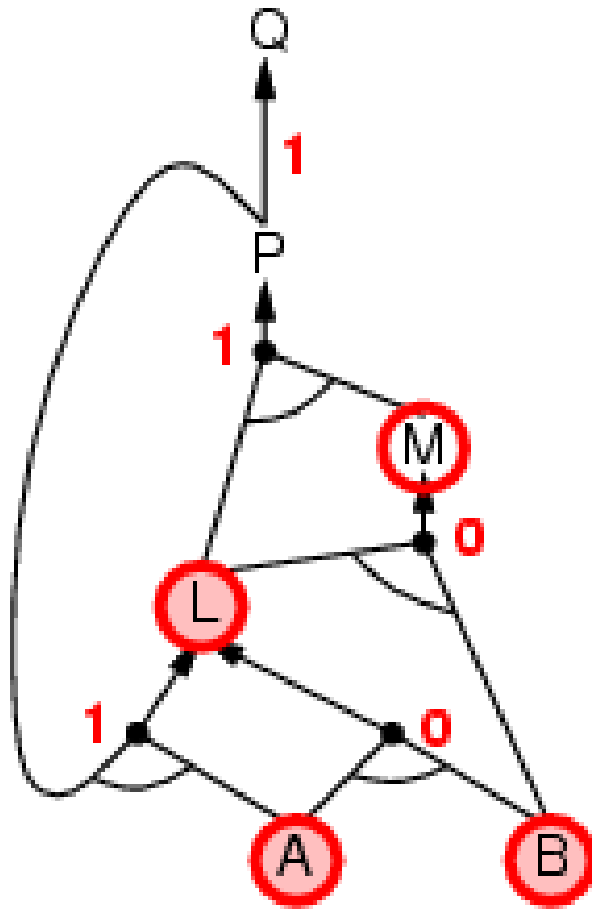
Forward chaining example



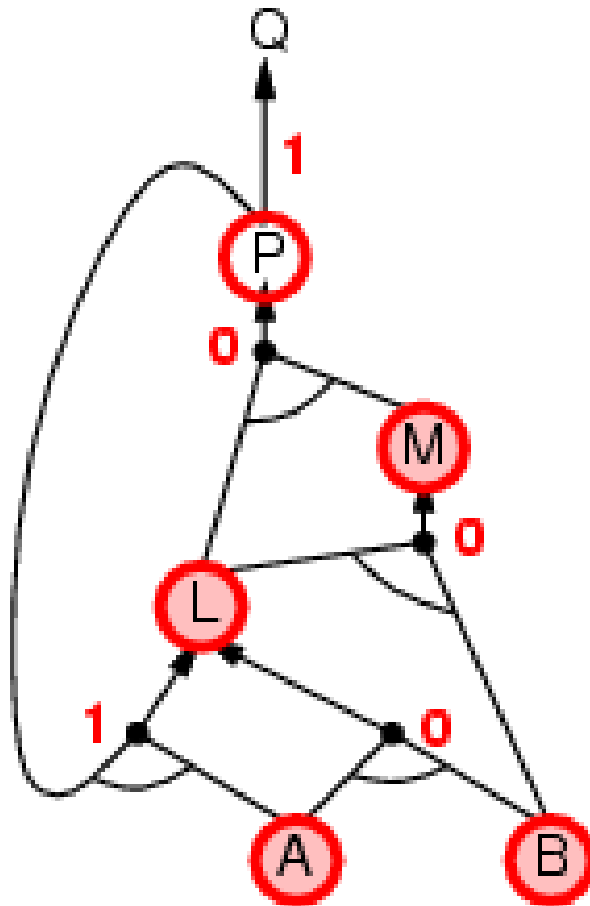
Forward chaining example



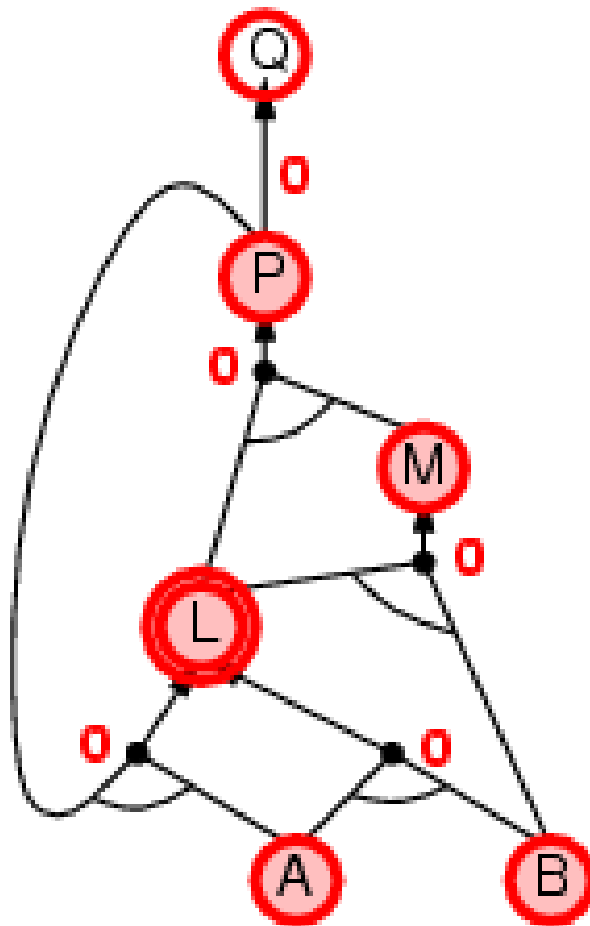
Forward chaining example



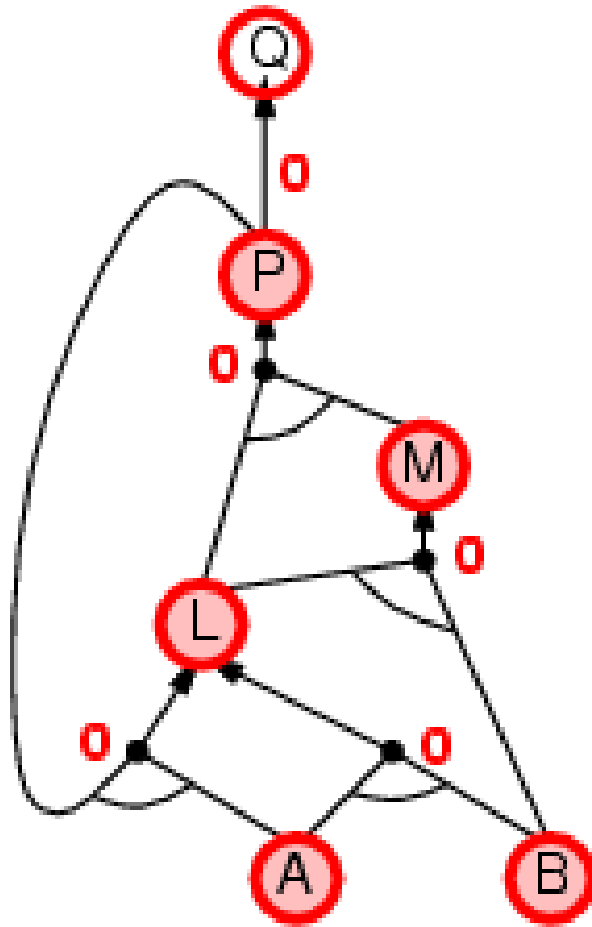
Forward chaining example



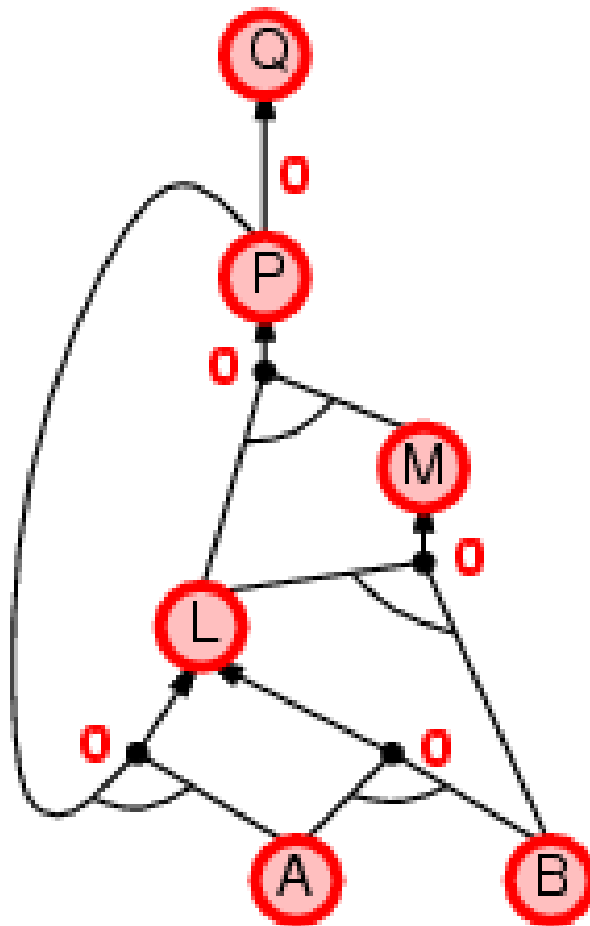
Forward chaining example



Forward chaining example



Forward chaining example



Proof of completeness

FC derives every atomic sentence that is entailed by KB

1. FC reaches a **fixed point** where no new atomic sentences are derived
2. Consider the final state as a model m , assigning true/false to symbols
3. Every clause in the original KB is true in m
$$a_1 \wedge \dots \wedge a_k \Rightarrow b$$
4. Hence m is a model of KB
5. If $KB \models q$, q is true in **every** model of KB , including m

Backward chaining

Idea: work backwards from the query q :

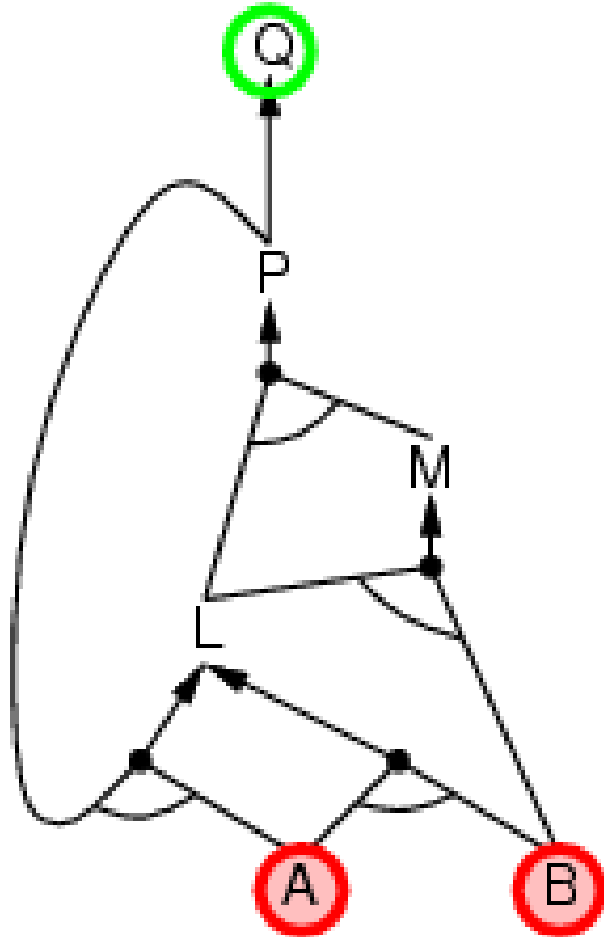
to prove q by BC,
check if q is known already, or
prove by BC all premises of some rule concluding q

Avoid loops: check if new subgoal is already on the goal stack

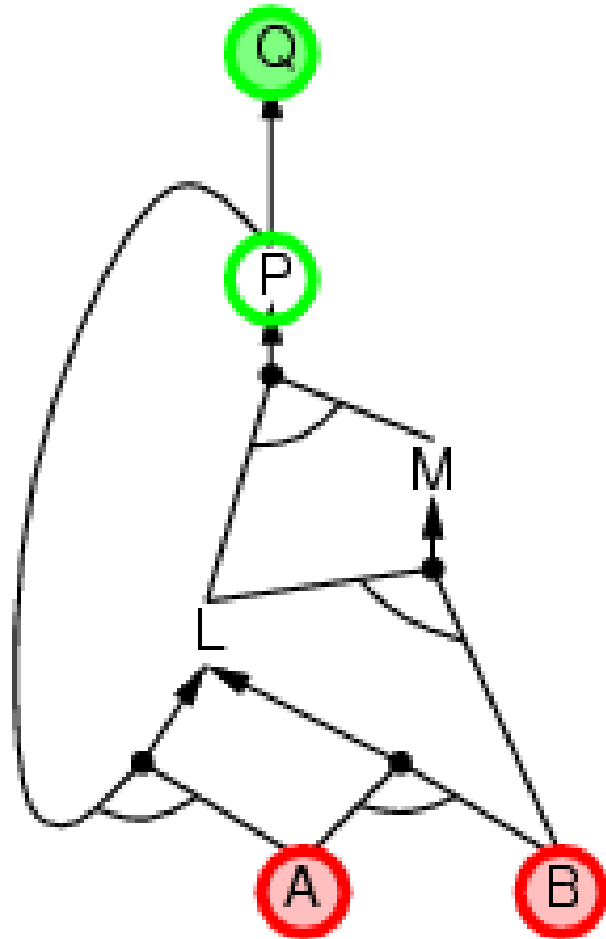
Avoid repeated work: check if new subgoal

1. has already been proved true, or
2. has already failed

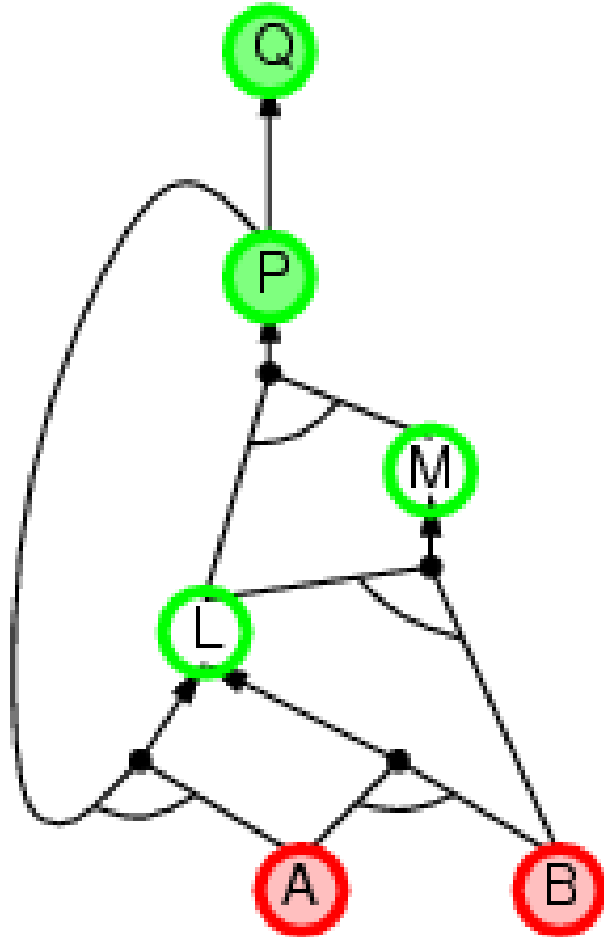
Backward chaining example



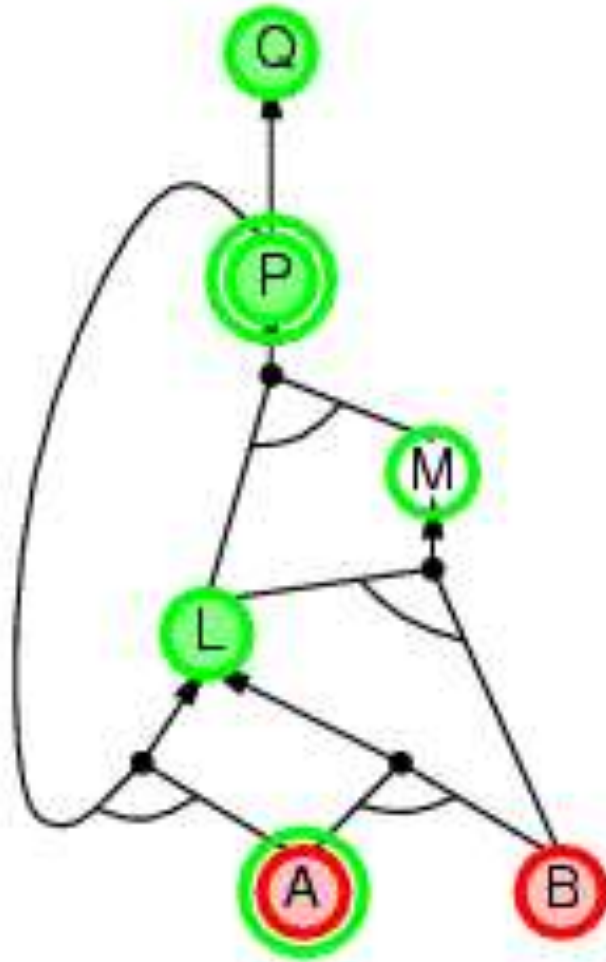
Backward chaining example



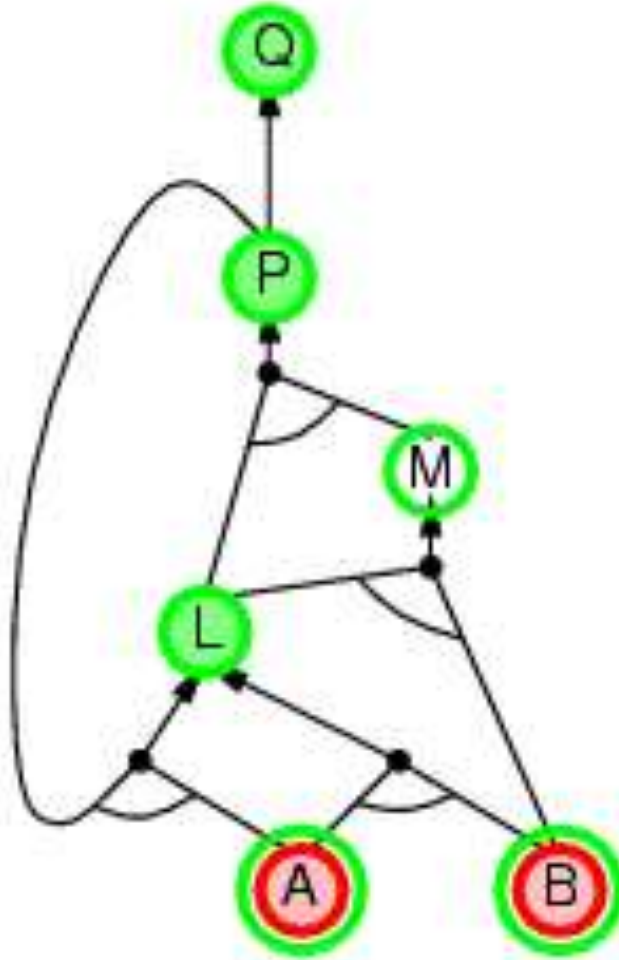
Backward chaining example



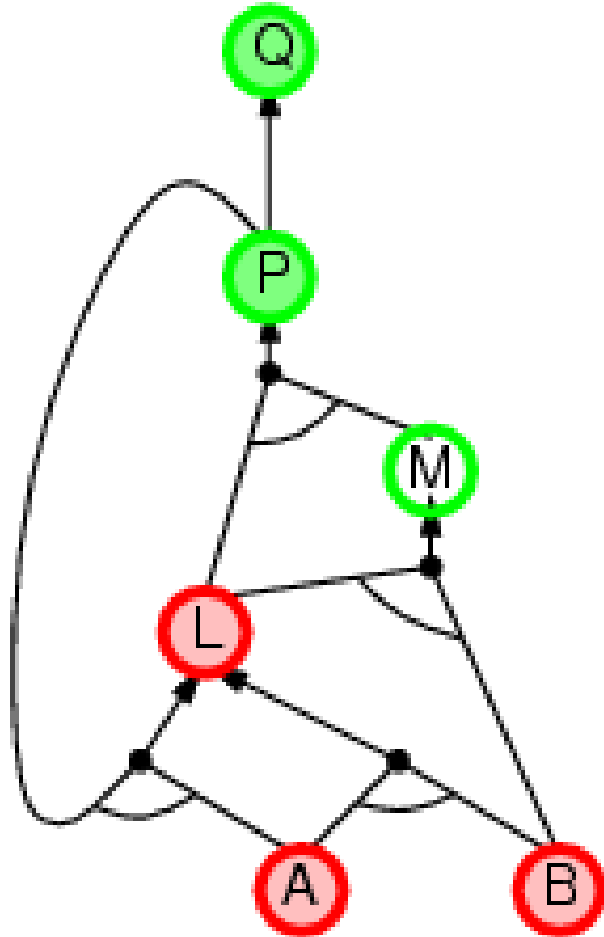
Backward chaining example



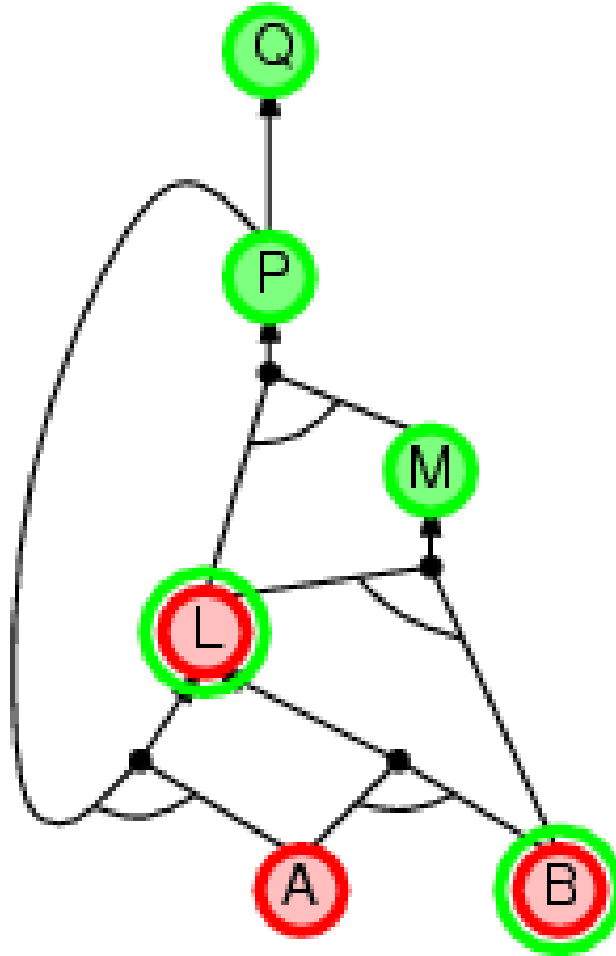
Backward chaining example



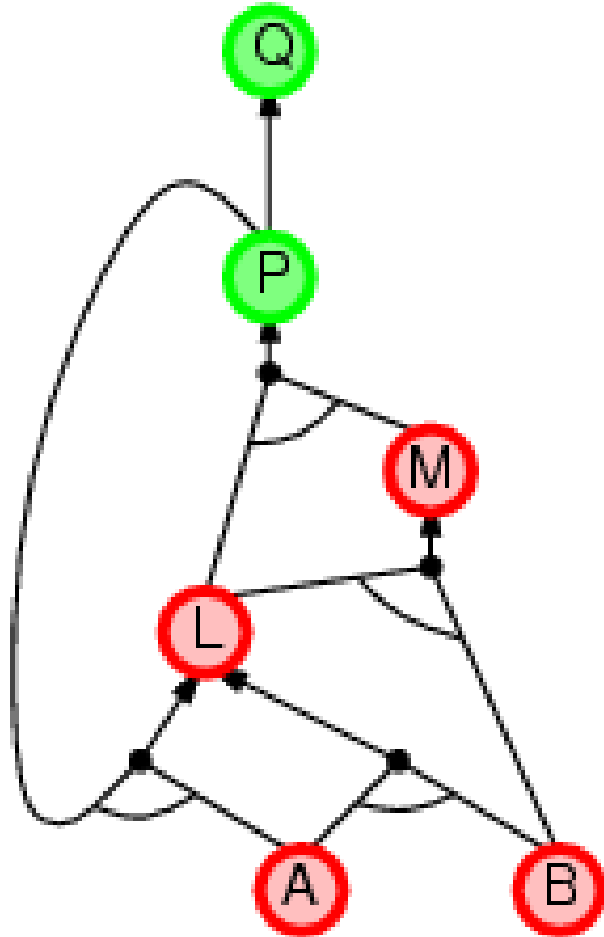
Backward chaining example



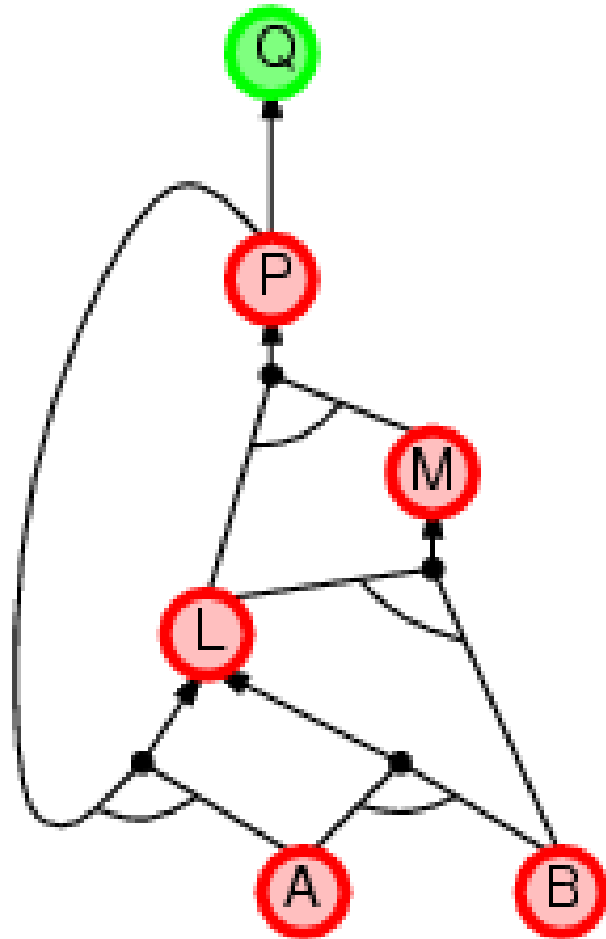
Backward chaining example



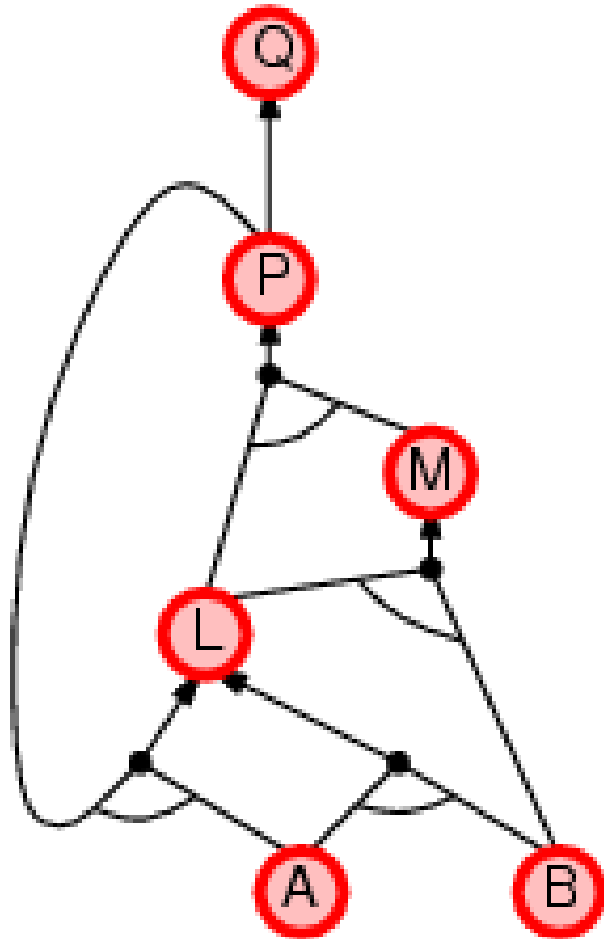
Backward chaining example



Backward chaining example



Backward chaining example



Forward vs. backward chaining

- FC is **data-driven**, automatic, unconscious processing,
 - e.g., object recognition, routine decisions
- May do lots of work that is irrelevant to the goal
- BC is **goal-driven**, appropriate for problem-solving,
 - e.g., Where are my keys? How do I get into a PhD program?
- Complexity of BC can be **much less** than linear in size of KB

Summary

- Logical agents apply **inference** to a **knowledge base** to derive new information and make decisions
- Basic concepts of logic:
 - **syntax**: formal structure of **sentences**
 - **semantics**: **truth** of sentences wrt **models**
 - **entailment**: necessary truth of one sentence given another
 - **inference**: deriving sentences from other sentences
 - **soundness**: derivations produce only entailed sentences
 - **completeness**: derivations can produce all entailed sentences
- Wumpus world requires the ability to represent partial and negated information, reason by cases, etc.
- Resolution is complete for propositional logic. Forward, backward chaining are linear-time, complete for Horn clauses
- Propositional logic lacks expressive power